

Topics in String Theory

M.Sc. Project Report

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She helped me in improving my report. Initially, there were lot of mistakes in the report like grammatical mistakes, too lengthy, unclear points, lack of motivation for topics.

Motivation

The modern problem in physics is quantum gravity. String theory tries to solve this problem. String in string theory has different vibrational modes just like ordinary strings and each vibrational mode corresponds a particle. One of the vibrational mode is graviton, a quantum particle of gravity. But this is not the end, general relativity is a classical theory and one cannot apply usual laws of quantum mechanics to general relativity and combine it with other three forces. String theory tries to solve this problem. Besides this string theory also attempts to address other questions in fundamental physics, cosmology, nuclear physics, black hole and early universe. In short string theory has connection to other areas of physics.

Current model of unification is standard model of physics which unifies all the forces except gravity. String theory supersedes standard model and tries to unify all forces and particle in a single mathematical framework.

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Chapter 1

Introduction

String theory allows more than three spacial dimensions ten in total. It turns out these extra dimensions are so compact that we cannot detect it experimentally. We will see one example of compact dimension in first chapter. One more interesting thing we note is that endpoints of a open string move with speed of light and actual physical motion of string is transverse to the string.

Conventional string end points have Neumann boundary conditions. However, putting Dirichlet boundary conditions on endpoints leads to the emergence of membranes and other hypersurfaces embedded in 9-dimensional space. Higher dimensional membrane is called a D-brane. D-branes are physical objects in string theory on which string endpoints are constrained to move.

In this first stage of my project, I have reviewed light-cone approach to study string theory. The plan of the report as follows: In chapter 2, first I will introduce light-cone coordinates and light-cone momentum then Lorentz invariant interval and metric for light-cone coordinates. Idea of compact dimension is discussed followed by Maxwell's equation in covariant form, from this we try to see how Gauss's remains invariant in higher dimensions. I also discussed how Einstein's gravitational equation under small perturbation on top of Minkowski metric will become. In third chapter we will see equation of motion for nonrelativistic strings, then I will derive action for relativistic point particle by using dimensions. By using this action, action for relativistic strings can be derived. In fourth chapter I will discuss various parametric equations satisfied by strings followed by one example. Various conserved current on world-sheet will also be discussed. In fifth chapter light-cone solution for relativistic string is derived and in the last relativistic point particle is discussed.

Chapter 2

Special Relativity, electromagnetism and gravitation in various dimensions

Since I have used light-cone approach to study string theory, I'd like to introduce light-cone coordinates, light-cone energy and review special relativity. As string theory includes all interaction, therefore we will see compact extra space dimensions followed by Maxwell electrodynamics and gravitation. We review the relativistic formulation of four-dimensional electrodynamics and show how it facilitates the definition of electrodynamics in other dimensions. Finally will see a brief description of Einstein's gravity.

We define the light-cone coordinates x^+ and x^- as linear combinations of the time coordinate x^0 and one spatial coordinate x^1 as follows

$$x^+ = \frac{(x^0 + x^1)}{\sqrt{2}}, \quad x^- = \frac{(x^0 - x^1)}{\sqrt{2}}. \quad (2.1)$$

While other coordinates are untouched. We can think of either of the light-cone coordinates as our new light-cone time. In our case we take x^+ to be the light-cone time and x^- as a spatial coordinate. Invariant interval ds^2 in light-cone coordinates can be written as

$$-ds^2 = -2dx^+dx^- + (dx^2)^2 + (dx^3)^2. \quad (2.2)$$

The matrix representation of the light-cone metric becomes

$$\eta_{\mu\nu} = \begin{pmatrix} 0 & -1 & 0 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (2.3)$$

Light-cone momentum coordinates p^+ and p^- are defined in same way as x^+ and x^-

$$p^+ = \frac{p^0 + p^1}{\sqrt{2}}, \quad p^- = \frac{p^0 - p^1}{\sqrt{2}}. \quad (2.4)$$

If we write Schrödinger equation in light-cone coordinates, it can be seen that x^+ gives rise to p^- and we know time gives rise to energy. Using this, it can be said that p^- is light-cone energy and p^+ is light-cone momentum.

Now turning to **compact extra dimension**. String theory has more than three dimensions but due to their compactness we can't see higher dimensions. Suppose you are walking and notice same scenery is repeating each time you move a distance $2\pi R$. There is no way to distinguish a line with such property from a circle with circumference $2\pi R$. In such cases, x coordinate of a line is said to be identified as

$$x \sim x + 2\pi R. \quad (2.5)$$

Next, to get some idea about higher dimensions we will see electromagnetism in four dimensions and extend it to higher dimensions. A four vector can be obtained by combining Φ and $\vec{A}$ as follows

$$A^\mu = (\Phi, A^1, A^2, A^3) \quad (2.6)$$

From A_μ we create *field strength* $F_{\mu\nu}$ as

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad (2.7)$$

Four Maxwell's equations can clubbed to two equations as

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \quad (2.8)$$

$$\frac{\partial F^{\mu\nu}}{\partial x^\nu} = \frac{1}{c} j^\mu \quad (2.9)$$

where $j^\mu = (c\rho, j^1, j^2, j^3)$. First one gives two source free equations and second one gives two equations with source terms. Maxwell's equations in this form are easily generalized to higher dimensions. For d spatial dimensions there are d $\vec{A}$'s and $\vec{j}$'s. Now $\vec{\nabla}$ and ∇^2 will also include d spatial derivative. Put $\mu = 0$ in (2.9), we get

$$\frac{\partial F^{0i}}{\partial x^i} = \frac{1}{c} \rho. \quad (2.10)$$

By using $F^{0i} = E_i$, above equation is Gauss's law in d spatial dimensions for $i = 1$ to d

$$\nabla \cdot \vec{E} = \rho. \quad (2.11)$$

In this way we can generalize all the equations. One more thing to is that Maxwell's equations are gauge invariant.

Now let us see equation **gravitational field** and its invariance. For weak gravitational field and in absence of source, Einstein's equations can be expanded in powers of $h_{\mu\nu}(x)$ as

$$\partial^2 h^{\eta\nu} - \partial_\alpha (\partial^\mu h^{\nu\alpha} + \partial^\nu h^{\mu\alpha}) + \partial^\mu \partial_\nu h = 0 \quad (2.12)$$

here $h_{\mu\nu}$ is small fluctuation about Minkowski metric and $h = \eta^{\mu\nu} h_{\mu\nu}$. Under gauge transformation

$$\delta_0 h^{\mu\nu} = \partial^\mu \epsilon^\nu + \partial^\nu \epsilon^\mu \quad (2.13)$$

(2.13) is invariant. We will use these equations in chapter 5 to find independent modes in fields.

Chapter 3

Nonrelativistic strings, relativistic point particle and relativistic strings

To appreciate relativistic strings, we will first see physics of nonrelativistic strings. Lagrangian, momentum density, equation of motion for nonrelativistic strings will be reviewed. Next comes relativistic string, we review action for relativistic particle and see how it naturally generalizes to action for relativistic string. We will focus on the surface traced out by the string in spacetime. We use the proper area of this surface as the action and it Nambu–Goto action. We also see the reparameterization property of this action, identify the string tension, and find the equations of motion. For open strings we see boundary conditions satisfied by string endpoints and introduce D-branes. Finally, we see the physical motion is transverse to the string.

We begin with classical strings. *Lagrangian density* for nonrelativistic string is given by

$$\mathcal{L} \left(\frac{\partial y}{\partial x}, \frac{\partial y}{\partial t} \right) = \frac{1}{2} \mu_0 \left(\frac{\partial y}{\partial t} \right)^2 - \frac{1}{2} T_0 \left(\frac{\partial y}{\partial x} \right)^2. \quad (3.1)$$

We define

$$\mathcal{P}^t = \frac{\partial \mathcal{L}}{\partial t}, \quad \mathcal{P}^x = \frac{\partial \mathcal{L}}{\partial x} \quad (3.2)$$

where

$$\mathcal{P}^t = \mu_0 \frac{\partial y}{\partial t}, \quad \mathcal{P}^x = -T_0 \frac{\partial y}{\partial x}. \quad (3.3)$$

Equation of motion for nonrelativistic string is

$$\frac{\partial^2 y}{\partial x^2} - \frac{1}{v_0^2} \frac{\partial^2 y}{\partial t^2} = 0 \text{ or } \frac{\partial \mathcal{P}^t}{\partial t} + \frac{\partial \mathcal{P}^x}{\partial x} = 0 \text{ with } v_0 = \sqrt{\frac{T_0}{\mu_0}} \quad (3.4)$$

We have two boundary conditions

$$\frac{\partial y}{\partial x}(t, x_*) = 0 \text{ Neuman boundary condition} \quad (3.5)$$

$$\frac{\partial y}{\partial t}(t, x_*) = 0 \text{ Dirichlet boundary condition} \quad (3.6)$$

x_* is boundary point. First condition is for free end point and second is for fixed end point.

We have to construct action for relativistic point particle, given a constraint that action is invariant. It can be done by using dimensions. Action has unit of energy times time

$$[s] = M \frac{L^2}{T^2} T. \quad (3.7)$$

We know proper time dt_p is Lorentz invariant. If we multiply dt_p with mc^2 which has unit of energy then the result will be action S . Note m is rest mass and it Lorentz invariant.

$$S = -mc \int ds, \quad dt_p = \frac{ds}{c} \quad (3.8)$$

Does the value of the action depend on the choice of parameter? Let us check. We now reexpress the integrand ds using the parameterized world-line. Earlier $ds = -\eta_{\mu\nu} dx^\mu dx^\nu$, now it becomes

$$ds^2 = -\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} (d\tau)^2. \quad (3.9)$$

For particle with velocity less than speed of light $ds = \sqrt{ds^2}$. Put this and (3.9) in (3.8) we get

$$S = -mc \int \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} d\tau \quad (3.10)$$

If we change parameter τ to τ' then

$$\frac{dx^\mu}{d\tau} = \frac{dx^\mu}{d\tau'} \frac{d\tau'}{d\tau}. \quad (3.11)$$

Put (3.11) in (3.10), S is

$$S = -mc \int \sqrt{-\eta_{\mu\nu} \frac{dx^\mu}{d\tau'} \frac{dx^\nu}{d\tau'}} d\tau'. \quad (3.12)$$

Which shows S is invariant of parameterization of world-line. This is important because one may say that I want to parameterize world-line differently, but result will be the same.

Area for spacetime surface which is parameterized by σ and τ is

$$A = \int d\sigma d\tau \sqrt{\left(\frac{\partial \vec{x}}{\partial \sigma} \cdot \frac{\partial \vec{x}}{\partial \sigma} \right) \left(\frac{\partial \vec{x}}{\partial \tau} \cdot \frac{\partial \vec{x}}{\partial \tau} \right) - \left(\frac{\partial \vec{x}}{\partial \sigma} \cdot \frac{\partial \vec{x}}{\partial \tau} \right)^2} \quad (3.13)$$

It is a proper area because it is Lorentz invariant. Area has unit of M^2 and if multiply by unit of M/T we will get unit of action. T_0/c has unit of M/T . Our action becomes

$$S = -\frac{T_0}{c} \int_0^{\sigma_1} d\sigma \int_{\tau_i}^{\tau_f} d\tau \sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2} \quad (3.14)$$

The above action is called **Nambu-goto action**. Note $x(\sigma, \tau) = X$ to simplify our equation. Lagrangian density is given by

$$\mathcal{L}(\dot{X}^\mu, X^{\mu'}) = -\frac{T_0}{c} \sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2}. \quad (3.15)$$

Conjugate momentum density similar to nonrelativistic strings are

$$\mathcal{P}_\mu{}^\tau = \frac{\partial \mathcal{L}}{\partial \dot{X}^\mu} = -\frac{T_0}{c} \frac{(\dot{X} \cdot X') X_\mu' - (X')^2 \dot{X}_\mu}{\sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2}} \quad (3.16)$$

$$\mathcal{P}_\mu{}^\sigma = \frac{\partial \mathcal{L}}{\partial X^{\mu'}} = -\frac{T_0}{c} \frac{(\dot{X} \cdot X')\dot{X} - (\dot{X})^2 X_\mu{}'}{\sqrt{(\dot{X} \cdot X')^2 - (\dot{X})^2 (X')^2}}. \quad (3.17)$$

Equation of motion in term of momentum density is

$$\frac{\partial \mathcal{P}_\mu{}^\tau}{\partial \tau} + \frac{\partial \mathcal{P}_\mu{}^\sigma}{\partial \sigma} = 0 \quad (3.18)$$

$$\textit{Dirichlet or fixed endpoint boundary condition} : \frac{\partial X^\mu}{\partial \tau}(\tau, \sigma_*) = 0 \quad (3.19)$$

$$\textit{free endpoint boundary condition} : \mathcal{P}_\mu{}^\sigma(\tau, \sigma_*) = 0, \sigma_* \textit{ is endpoint} \quad (3.20)$$

The objects on which open string endpoints must move are characterized by the number of spatial dimensions that strings have. They are called **D-branes**, where the letter D stands for Dirichlet. A Dp -brane is an object with p spatial dimensions. If a set of Dirichlet boundary conditions is specified, then string is guaranteed to move in this Dp brane. A flat $D2$ brane in a three dimensional space which is specified by one condition, let say $x^3 = 0$. This means that the $D2$ brane is just (x^1, x^2) plane.

Two important things that can be derived are

- The endpoints move with the speed of light.
- The endpoints move transversely to the string.

Chapter 4

String parameterization, classical motion and world-sheet currents

We construct lines of constant σ that are perpendicular to the lines of constant τ to fix the σ parameterization. We will be left string equations of motion which is wave equations and two constraints. We find the solution which is going the equation of motion for open strings with free endpoints. Next by using symmetry we try to find conserved current and see that in string theory there are current that flow on two dimensional world-sheet which is traced by string in spacetime. Conserved charges associated with these conserved governs the motion of string. We give a simple physical interpretation of the objects $\mathcal{P}(\tau)$ and $\mathcal{P}(\sigma)$ which we encountered earlier.

We have four equations

$$\text{wave equation : } \frac{\partial^2 \vec{X}}{\partial \sigma^2} - \frac{1}{c^2} \frac{\partial^2 \vec{X}}{\partial t^2} = 0 \quad (4.1)$$

$$\text{parameterization equation : } \frac{\partial \vec{X}}{\partial t} \cdot \frac{\partial \vec{X}}{\partial \sigma} = 0 \quad (4.2)$$

$$\text{parameterization : } \left(\frac{\partial \vec{X}}{\partial \sigma} \right)^2 + \frac{1}{c^2} \left(\frac{\partial \vec{X}}{\partial t} \right)^2 = 1 \quad (4.3)$$

$$\text{boundary condition : } \left. \frac{\partial \vec{X}}{\partial \sigma} \right|_{\sigma=0} = \left. \frac{\partial \vec{X}}{\partial \sigma} \right|_{\sigma=\sigma_1} = 0. \quad (4.4)$$

Note equations are written in static gauge $\tau = t$. (4.2) tells that motion is transverse to string. (4.4) is free endpoint condition with endpoints at $\sigma = 0$ and $\sigma = \sigma_1$. Let us first consider the wave equation for $\vec{X}$. This equation is readily solved in terms of arbitrary vector functions of $(ct \pm \sigma)$. We thus write

$$\vec{X}(ct, \sigma) = \frac{1}{2} (\vec{F}(ct + \sigma) + \vec{G}(ct - \sigma)). \quad (4.5)$$

Using (4.2),(4.3),(4.4) in (4.5) we get solution for open string as

$$\vec{X}(ct, \sigma) = \frac{1}{2} (\vec{F}(ct + \sigma) + \vec{F}(ct - \sigma)) , \quad \sigma \in (0, \sigma_1) \quad (4.6)$$

with constraints on $\vec{F}$ as follows

$$\left| \frac{\partial \vec{F}(u)}{\partial u} \right|^2 = 1 , \quad \vec{F}(u + 2\sigma_1) = \vec{F}(u) + 2\sigma_1 \frac{\vec{v}_0}{c} \quad (4.7)$$

v_0 is like average velocity over span of $2\sigma_1$. Let us study one case. Consider an open string of length l rotating with fixed center such that it has $l/2$ fraction on both the sides and angular frequency ω . We know energy E of stretch with tension T_0 and length σ_1 is

$$E = T_0 \sigma_1 \quad (4.8)$$

in this case string has only potential energy because it is only stretched and has no kinetic energy. Energy density for this case is

$$\varepsilon = T_0. \quad (4.9)$$

We $\sigma = 0$ endpoint as

$$\vec{X}(t, 0) = \frac{l}{2}(\cos(\omega t), \sin(\omega t)) \quad (4.10)$$

Given that $\vec{F}(ct) = \vec{X}(ct, 0)$ we have

$$\vec{F}(u) = \frac{l}{2} \left(\cos\left(\frac{\omega u}{c}\right), \sin\left(\frac{\omega u}{c}\right) \right). \quad (4.11)$$

Since $\vec{v}_0$ is average velocity over one cycle, in this case due to circular motion $\vec{v}_0 = 0$. Therefore second equation of (4.7) becomes $\vec{F}(u + 2\sigma_1) = \vec{F}(u)$. Using this in (4.9) we get

$$\frac{\omega}{c}(2\sigma_1) = 2\pi m \rightarrow \frac{\omega}{c} = \frac{\pi}{\sigma_1} m \quad (4.12)$$

we must choose $m = 1$, otherwise $\vec{X}(0, \sigma)$ solution of at $t = 0$ for $\sigma \in [0, \sigma_1]$ will trace string m times. For $m = 1$

$$\frac{\omega}{c} = \frac{\pi}{\sigma_1} = \frac{\pi T_0}{E} \quad (4.13)$$

for last term (4.8) is used. Now using first condition in (4.7) and (4.10) we get

$$\left| \frac{d\vec{F}}{du} \right|^2 = \left(\frac{\omega l}{2c} \right)^2 = 1 \rightarrow l = \frac{2c}{\omega} = \frac{2\sigma_1}{\pi} = \frac{2E}{\pi T_0} \quad (4.14)$$

for last two terms (4.12) is used. Energy E becomes

$$E = \frac{\pi}{2} T_0 l \quad (4.15)$$

energy of just stretch was $T_0 l$ now it includes kinetic also. $\vec{F}(u)$ in terms of σ_1

$$\vec{F}(u) = \frac{\sigma}{\pi} \left(\cos\left(\frac{\omega u}{c}\right), \sin\left(\frac{\omega u}{c}\right) \right) \quad (4.16)$$

put this in (4.6) to get

$$\vec{X}(ct, \sigma) = \frac{\sigma_1}{\pi} \cos \frac{\pi \sigma}{\sigma_1} \left(\cos \frac{\pi ct}{\sigma_1}, \sin \frac{\pi ct}{\sigma_1} \right) \quad (4.17)$$

energy $\varepsilon(s)$ per unit length on the string as a function of the distance s to the center:

$$\varepsilon(s) = \frac{T_0}{\sqrt{1 - \frac{4s^2}{l^2}}}. \quad (4.18)$$

At the center of the string the energy density $\varepsilon(0)$ coincides with T_0 . This had to be so, because the string does not move at the center (see (4.9)). The energy density diverges at the string endpoints $s = \pm l/2$. The total energy, however, is finite.

Next, we will see conserved current and conserved charges, first by constant infinitesimal transformation and second by Lorentz symmetry. Action is given by

$$S = \int d\xi^0 d\xi^1 \mathcal{L}(\partial_0 X^\mu, \partial_1 X^\mu) \quad (4.19)$$

with $(\xi^0, \xi^1) = (\tau, \sigma)$ (see (3.15)). We note that infinitesimal constant transformation

$$\delta X^\mu(\tau, \sigma) = \epsilon^\mu \quad (4.20)$$

leaves $\mathcal{L}(\partial_0 X^\mu, \partial_1 X^\mu)$ invariant since Lagrangian only involves derivative terms. Now conserved current for such transformation is

$$j^\alpha{}_\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\alpha X^\mu)} \quad (4.21)$$

for $\alpha = 1, 2$ we get

$$(j^0{}_\mu, j^1{}_\mu) = \left(\frac{\partial \mathcal{L}}{\partial \dot{X}^\mu}, \frac{\partial \mathcal{L}}{\partial X^{\mu'}} \right) = (\mathcal{P}_\mu{}^\tau, \mathcal{P}_\mu{}^\sigma) \quad (4.22)$$

first two terms are just index expansion and the last we got from (3.16) and (3.17). From above equation it seen that j 's are momentum density $\mathcal{P}_\mu{}^\alpha$ defined in chapter 3. Integrate j^0 component over σ to get conserved current as

$$p_\mu(\tau) = \int_0^{\sigma_1} j^0{}_\mu d\sigma = \int_0^{\sigma_1} \mathcal{P}_\mu{}^\tau d\sigma. \quad (4.23)$$

Note this definition of charge is familiar from electromagnetism, we integrate $j^0 = \rho$ to get electric charge. $\mathcal{P}_\mu{}^\tau$ is the σ density of spacetime momentum carried by the string. Conservation of p_μ can be checked as follows

$$\frac{dp_\mu}{d\tau} = \int_0^{\sigma_1} \frac{\partial \mathcal{P}_\mu{}^\tau}{\partial \tau} d\sigma = - \int_0^{\sigma_1} \frac{\partial \mathcal{P}_\mu{}^\sigma}{\partial \sigma} d\sigma = - \int_0^{\sigma_1} \mathcal{P}_\mu{}^\sigma d\sigma = 0 \quad (4.24)$$

from second to third term we used (3.18) and from fourth to fifth term we used (3.20). Now let's see conserved current and charges for transformation $\delta X^\mu = \epsilon^{\mu\nu} X_\nu$ such that it leaves Lagrangian invariant. ϵ is infinitesimal constant matrix element. We can show that if Lagrangian Lorentz invariant, then ϵ has to be antisymmetric. By using general for conserved current similar to (4.21) we get conserved current as

$$\mathcal{M}_{\mu\nu}{}^\alpha = X_\mu \mathcal{P}_\nu{}^\alpha - X_\nu \mathcal{P}_\mu{}^\alpha \quad (4.25)$$

Corresponding conserved charges are

$$M_{\mu\nu} = \int (\mathcal{M}_{\mu\nu}{}^\tau d\sigma - \mathcal{M}_{\mu\nu}{}^\sigma d\tau). \quad (4.26)$$

M^{0i} charges are associated with boost and the components M^{ij} measure the string angular momentum $\vec{L}$ via the usual relations

$$L_i = \frac{1}{2} \epsilon_{ijk} M_{jk}. \quad (4.27)$$

Chapter 5

Light-cone relativistic strings, fields and particles and the relativistic quantum point particle

In this chapter we introduce a class of gauges is introduced which fix the parameterization of the world-sheet leading to equations of constraints, and give equations of motion that are wave equations. We will see in the light-cone gauge, sets X^+ proportional to τ and allows for a solution to the equations of motion. The dynamics of the string in this gauge is determined by the motion along the transverse directions and the values of two zero modes. We see the classical Virasoro modes as the oscillation modes of the X^- coordinate. We see how to calculate the mass of an arbitrary string configuration. Next, equations of motion for scalar fields, Maxwell fields, and gravitational fields is review classically. We use the light-cone gauge to find plane-wave solutions to their equations of motion and the number of degrees of freedom present in them. Then by quantization of these fields we get scalar particle, photons and gravitons. In this way we identify of such states among the quantum states of relativistic strings. We will see that the quantum states of the relativistic point particle coincide with the one-particle states of the quantum scalar field. We will see translation generator and Lorentz transformation generator.

In this chapter τ and σ are unitless. Natural units are used $\hbar = 1$ and $c = 1$. Gauge for τ and parameterization for σ gives

$$n \cdot X(\tau, \sigma) = \beta \alpha' (n \cdot p) \tau \quad (5.1)$$

$$n \cdot p = \frac{2\pi}{\beta} n \cdot \mathcal{P}^\tau \quad (5.2)$$

$$n \cdot \mathcal{P}^\sigma = 0 \quad (5.3)$$

$$\beta = 2 \text{ for open string}$$

$$\beta = 1 \text{ for closed string}$$

α' is constant. Above equations are valid for both open and closed strings. These are the parameterization in which we to find equation of motion and it's solution for strings. By using (5.1),(5.3),(3.16) and (3.17) we get

$$\dot{X} \cdot X' = 0, \quad \dot{X}^2 + X'^2 = 0. \quad (5.4)$$

Put this back in (3.16) and (3.17) to get momentum density as

$$\mathcal{P}^{\tau\mu} = \frac{1}{2\pi\alpha'} \dot{X}^\mu, \quad \mathcal{P}^{\sigma\mu} = \frac{-1}{2\pi\alpha'} X'^\mu \quad (5.5)$$

Using continuity equation we write

$$\partial_\tau \mathcal{P}^{\tau\mu} + \partial_\sigma \mathcal{P}^{\sigma\mu} = 0 \quad (5.6)$$

note this is the same continuity equation as electromagnetism one $\partial_\mu j^\mu = 0$. From above two equations we get equation of motion

$$\ddot{X}^\mu - X^{\mu''} = 0 \quad (5.7)$$

which is wave equation.

$\vec{X}$ can be expressed in terms of $\vec{F}$ (see (4.4)). By using boundary condition (4.4) it can be shown $\vec{F}'$ is periodic. Now $\vec{X}$ can be written in terms of sin and cos as follows

$$X^\mu = x_0^\mu + \sqrt{2\alpha'}\alpha_0^\mu\tau + i\sqrt{2\alpha'}\sum_{n \neq 0} \frac{1}{n}\alpha_n^\mu e^{-in\tau} \cos(n\sigma) \quad (5.8)$$

here α' is slope parameter which relates angular momentum and energy. The terms on the right-hand side clearly correspond to the zero mode, to the momentum, and to the oscillations of the string. If all the coefficients α_n^μ of the oscillations vanish, the equation represents the motion of a point particle.

Next we define *transverse Virasoro* mode $L_n^\perp$ and *mass* of string M as follows

$$L_n^\perp = \frac{1}{2} \sum_{p \in \mathbb{R}} \alpha_{n-p}^I \alpha_p^I \quad p \in \mathbb{R} \quad (5.9)$$

$$M^2 = \frac{1}{\alpha'} \sum_{n=1}^{\infty} n a_n^{I*} a_n^I \quad (5.10)$$

where α_n in (5.10) is related to a_n

$$\alpha_n^\mu = a_n^\mu \sqrt{n}, \quad \alpha_{-n}^\mu = a_n^{\mu*} \sqrt{n}. \quad (5.11)$$

Instead of writing X^- in terms of α_n , we can write it in terms of $L_n^\perp$.

Next, turning to light-cone fields and particles Equation of motion for scalar field is

$$(\partial^2 - m^2)\phi. \quad (5.12)$$

Classical plane wave solution for this is

$$\phi(t, \vec{x}) = a e^{-iEt + i\vec{p} \cdot \vec{x}}. \quad (5.13)$$

When we quantize it, ϕ becomes quantum scalar field operator as follows

$$\phi(t, \vec{x}) = \frac{1}{V} \sum_{\vec{p}} \frac{1}{\sqrt{2E_p}} (a_p e^{-iE_p t + i\vec{p} \cdot \vec{x}} + a^\dagger e^{iE_p t - i\vec{p} \cdot \vec{x}}) \quad (5.14)$$

where a_{-p} , a_p are annihilation operator and $a^\dagger$, $a^{-\dagger}$ are creation operator with commutation property

$$[a_p, a_p^\dagger] = 1, \quad [a_{-p}, a_{-p}^\dagger] = 1. \quad (5.15)$$

Vacuum state is defined as

$$a_p |\Omega\rangle = 0. \quad (5.16)$$

Similar to this multiparticle state

$$a_{p_1}^\dagger a_{p_2}^\dagger \dots a_{p_n}^\dagger |\Omega\rangle \quad (5.17)$$

and if we keep only one p , suppose p_1 then above equation becomes one particle state. Similarly we can write one particle state in light-cone as

$$a_{p+p^T}^\dagger |\Omega\rangle \quad (5.18)$$

p^T represents transverse momentum.

We now turn to an analysis of Maxwell fields and their corresponding quantum states. In absence of source (2.10) becomes

$$\partial_\nu(\partial^\mu A^\nu - \partial^\nu A^\mu) = 0. \quad (5.19)$$

This equation is invariant under gauge transformation

$$\delta A_\mu(p) = ip_\mu \epsilon(p) \quad (5.20)$$

note this equation is written in momentum space. Under light-cone gauge we try set

$$A^+ = 0 \quad (5.21)$$

and find independent modes. Gauge transformation for A^+ is

$$A(p)^{+'} = A(p)^+ + ip^+ \epsilon(p) \quad (5.22)$$

by adjusting ϵ we can make $A^+ = 0$. Our equation (10.8) becomes

$$A^- = \frac{1}{p^+} (p^I A^I) \quad (5.23)$$

where $I = 2 \text{ to } D$, for D spacetime dimensions, note A^- is not independent. From (5.21) and (5.23), we can determine motion of charged particle only by $(D-2)$ A^I . Therefore there are $(D-2)$ degrees of freedom or $(D-2)$ oscillators. One Photon state is superposition of $(D-2)$ oscillators

$$\sum_{I=2}^{D-1} \xi_I a_{p^+, p^T}^\dagger |\Omega\rangle \quad (5.24)$$

ξ_I is called polarization vector. Similarly we do for gravitation.

Next, let's see relativistic quantum point particle. In quantum theory variables become operator. Commutator relation becomes

$$[x^I, p^I] = i\eta^{IJ} \quad , \quad [x_0^-, p^+] = i\eta^{-+} = -i \quad (5.25)$$

What happens to conserved charges in the quantum theory? They become quantum operators with a remarkable property: they generate, via commutation, a quantum version of the symmetry transformation that gave rise to them classically! To see whether **momentum generates translation**, we check whether $i\epsilon_\rho p^\rho(\tau)$ generates translation

$$\delta x^\mu(\tau) = [i\epsilon_\rho p^\rho(\tau), x^\mu(\tau)] = i\epsilon(-i\eta^{\rho\mu}) = \epsilon^\mu \quad (5.26)$$

and it does. This is very elegant result.

Now let's **light-cone Lorentz generator**. Infinitesimal Lorentz generator of a point particle is

$$\delta^\mu(\tau) = \epsilon^{\mu\nu} x_\nu(\tau) \quad (5.27)$$

where $\epsilon^{\mu\nu} = -\epsilon^{\nu\mu}$. Associated charges are

$$M^{\mu\nu} = x^\mu(\tau)p^\nu(\tau) - x^\nu(\tau)p^\mu(\tau) \quad (5.28)$$

these charges are conserved classically. The quantum charges are expected to generate Lorentz transformation of the coordinates. Let's check

$$\delta x^\rho(\tau) = [-\frac{i}{2}\epsilon_{\mu\nu} M^{\mu\nu}, x^\rho(\tau)] = \frac{1}{2}\epsilon_{\mu\nu}(\eta^{\mu\rho}x^\nu(\tau) - \eta^{\nu\rho}x^\mu(\tau)) = \epsilon^{\rho\nu}x_\nu(\tau) \quad (5.29)$$

and $M^{\mu\nu}$ generates Lorentz transformation (5.27).

There is a natural identification of the quantum states of relativistic point particle of mass m with the one particle states of the a scalar field of mass m

$$|p^+, \vec{p}_T\rangle \leftrightarrow a_{p^+, p_T} |\Omega\rangle. \quad (5.30)$$

We can show, their equation of motion is also same.

Problem

Consistency checks for X^- coordinate.

1. We show that integrating dX^- from a to b , result is independent if X^I satisfies wave equation.
2. If transverse coordinate X^I , $I = 2, \dots, D$ satisfies wave equation then the X^- also satisfies wave equation.
3. By assuming at open string endpoints some of the transverse coordinates satisfies Dirichlet boundary condition and other transverse coordinates satisfies Neumann boundary conditions. We will prove X^- always satisfies Neumann boundary conditions.

Solution: 1. Using equation (5.4) to write $(\dot{X} \pm X')^2 = 0$, which further can be expanded in light-cone coordinates as follows

$$-2(\dot{X}^+ \pm X^{+'})(\dot{X}'^- \pm X'^-) + (\dot{X}^I \pm X'^I)^2 = 0 \quad (5.31)$$

We use light-cone gauge to solve the problem. In light-cone gauge we have

$$n_\mu = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0, \dots, 0 \right) \quad (5.32)$$

put this in (5.1) we get

$$X^+ = \beta \alpha' p^+ \tau. \quad (5.33)$$

Now differentiate (5.33) with respect to τ , σ and substitute in (5.31) to get

$$\dot{X}^- + X'^- = \frac{1}{4\alpha' p^+} (\dot{X}^I + X'^I)^2 \dot{X}^- - X'^- = \frac{1}{4\alpha' p^+} (\dot{X}^I - X'^I)^2 \quad (5.34)$$

add and subtract above two equations to get

$$\dot{X}^- = \frac{1}{2\alpha' p^+} (\dot{X}^{I^2} + X'^{I^2}), \quad X'^- = \frac{1}{4\alpha' p^+} (4\dot{X}^I X'^I). \quad (5.35)$$

Differentiate first equation of (5.35) with σ , second equation with τ and we find that

$$\partial_\sigma (\partial_\tau X^-) = \partial_\tau (\partial_\sigma X^-) \quad (5.36)$$

only if X^I satisfies wave equation. By using (5.36) we can write

$$dX^- = \frac{\partial X^-}{\partial \sigma} d\sigma + \frac{\partial X^-}{\partial \tau} d\tau = \int_a^b \nabla X^- \cdot d\vec{r} = X^-(b) - X^-(a) \quad (5.37)$$

(5.36) is used for two terms and last two terms are just gradient theorem. Above equation shows integration of dX^- is independent of path.

2. Differentiate first equation of (5.35) with τ and second equation by σ we find

$$\frac{\partial^2 X^-}{\partial \tau^2} = \frac{\partial^2 X^-}{\partial \sigma^2} \quad (5.38)$$

only if X^I satisfies wave equation.

3. Let some transverse coordinates satisfy Dirichlet boundary condition and some satisfy Neumann boundary condition, then we have $\dot{X}^I = 0$ and $X'^I = 0$ respectively at endpoints according to (3.5) and (3.6). Their multiplication will always be zero at endpoints, consequently from second equation of (5.35) we say $X'^- = 0$ at endpoints. Which shows that $X^- = 0$ will always satisfy Neumann boundary condition.

Chapter 6

Conclusion

First I reviewed special relativity and electromagnetism. Then I studied nonrelativistic strings then relativistic strings then went to light-cone relativistic strings. Parallely I did particle and field counterpart like relativistic point particle, light-cone fields and particles and relativistic quantum point particle. I have solve more than 40 problem from eleven chapters of Barton Zwiebach. Few problems are half solved and rest I am still trying to do. Idea behind doing project in introductory string theory is that I will get know to few basic things about it like string endpoints move speed of light, actual motion of the string is transverse to the string itself, strings have different vibrational mode called virasoro mode and many more.

Next semester I'd like to read research papers and I will parallely continue reading Barton Zwiebach. I have followed light-cone approach to study string theory next semester I try to study other approach and see how it is different from light-cone method.